INCH-POUND

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MILITARY SPECIFICATION

TEST SET, ANTENNA AN/VPM-1

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers antenna test set, AN/VPM-1, a rack-mounted test set for testing the elevation and traverse servo drive assemblies of Antenna AS-2049/VPS-2, part of Radar Set, AN/VPS-2 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines
MIL-STD-2000	-	Standard Requirements for Soldered Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10554576	-	Test Set, Antenna AN/VPM-1

PACKAGING DATA SHEET

P10554576	-	Packaging of Test Set, Antenna AN/VPM-1

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

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3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 10554576, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the antenna test set shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances
b. Inorganic protective surface finishings
c. Part identification and marking
d. Workmanship

3.3 Environmental service conditions.

3.3.1 Vibration. The antenna test set shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
5 to 15 hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 55 hertz	0.02 inches double amplitude

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ±10% per decade.
The 5-55-5 hertz sweep time is 7 minutes 35 seconds ±10%.

3.3.2 Shock. The antenna test set shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ±10%, 11 milliseconds ±10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude.

3.3.3.1 Operating. The antenna test set shall meet the requirements of 3.5.1.1 rated load only, 3.5.1.2 no load only, 3.5.1.3, 3.5.1.4 and 3.5.1.5 during and all the requirements of 3.4 and 3.5 after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.3.2 Non-operating. The antenna test set shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).

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3.3.4 Temperature.

3.3.4.1 Operating. The antenna test set shall meet the requirements of 3.5.1.1 rated load only, 3.5.1.2 no load only, 3.5.1.3, 3.5.1.4 and 3.5.1.5 when stabilized thermally at the temperature extremes of +32°F and +131°F.

3.3.4.2 Non-operating (storage). The antenna test set shall not suffer permanent damage when subjected to the constant temperature extremes of -60°F for a period of 24 hours and +160°F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The antenna test set shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical.

3.4.1 Continuity. Each conductor shall be electrically continuous as specified on the wire runs list.

3.4.2 Resistance. Resistance between front panel test points shall be as follows:

CX-R1 (TP1) to CX-R2 (TP2) - between 275 and 400 ohms

CX-S1 (TP3) to CX-S2 (TP4) - between 265 and 465 ohms

CX-S2 (TP4) to CX-S3 (TP5) - between 265 and 465 ohms

CX-S1 (TP3) to CX-S3 (TP5) - between 265 and 465 ohms

3.5 Performance. With input power applied, the antenna test set shall operate as follows:

3.5.1 Voltages. With the input voltage at 115 VAC ±10%, 400 Hz ±5% and with the appropriate switches turned on, the antenna test set shall produce the voltages given below, between the indicated terminals.

3.5.1.1 Brake voltage - no load and rated load. A fixed voltage of +24 ±2 Vdc with a load current capability ~~of 150 mA at +22~~ Vdc shall be available between terminals J1-a(Ret) and J1-n and between test points TP12 and TP13(Ret).

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3.5.1.2 Motor voltage - no load and rated load. Under no load, a continuously variable voltage from -19 ±2 Vdc to +19 ±2 Vdc shall be available between terminals J1-m(ret) and J1-r and between test points TP8 and TP9(Ret). Voltage under rated load shall be a maximum of ±9 Vdc.

3.5.1.3 $\pm KI^2M$ voltage - no load and rated load. A fixed voltage of +15 ±1.2 Vdc with a load current capability of 40 mA shall be available between terminals J1-h(Ret) and J1-E, -N, and between test points TP17 and TP18(Ret).

3.5.1.4 KI^2M voltage - no load and rated load. A fixed voltage of -15 ±1.2 Vdc with a load current capability of 40 mA shall be available between J1-h(Ret) and J-1F, -M, -f, -g, and between TP18(Ret) and TP19.

3.5.1.5 Synchro signal. A 3-wire synchro signal locked at electrical 0° shall be available at J1-c, J1-H and J1-G and at TP3, TP4 and TP5.

3.5.2 Circuit breakers. POWER circuit breaker CB1 and MOTOR circuit breaker CB2 shall be closed when reset button is pushed in so white band is concealed. Each circuit breaker shall be open when reset button is pulled out to expose white band, and shall open automatically within the times indicated below when the indicated current overload is passed through it:

Circuit Breaker	Overload Current	Time Limit
CB1	1.5 to 3.0 A dc	20 seconds
CB2	5 to 10 A dc	30 seconds

3.5.3 Circuit breaker CB1. With circuit breaker CB1 open, no indicators shall be illuminated and none of the output voltages of 3.5.1 shall appear at any of the output terminals.

3.5.4 POWER switch S1. With circuit breaker CB1 closed, POWER switch S1 shall serve to change indicator DS1 from OFF to ON and ON to OFF. With DS1 indicating OFF, none of the other indicators shall be illuminated.

3.5.5 SYNCHRO EXCITATION switch S2. With DS1 indicating ON, SYNCHRO EXCITATION switch S2 shall serve to change indicator DS2 from OFF to ON and ON to OFF, and to apply or disconnect the synchro signal of 3.5.1.5.

3.5.6 MOTOR TEST switch S3. With DS1 indicating ON, MOTOR TEST switch S3 shall serve to illuminate and extinguish the MOTOR TEST part of the indicator DS3.

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3.5.7 BRAKE RELEASED switch S5 and circuit breaker CB2. With the MOTOR TEST portion of indicator DS3 illuminated, BRAKE RELEASED switch S5 shall serve to illuminate and extinguish both the BRAKE RELEASED indicator DS5 and the ON part of the indicator DS3. With BRAKE RELEASED indicator DS5 illuminated, circuit breaker CB2 shall serve to apply (when closed) the voltage of 3.5.1.2 to the appropriate terminals, the voltage of 3.5.1.1 shall appear at the appropriate terminals, and either the CLOCKWISE or the CCW part of indicator DS4 shall be illuminated.

3.5.8 CLOCKWISE - CCW switch S4 and MOTOR VOLTAGE control T1. With BRAKE RELEASED indicator DS5 illuminated, CLOCKWISE-CCW switch S4 shall serve to change indicator DS4 from CLOCKWISE to CCW etc., and to preserve the polarity of the GEN OUTPUT test points (TP10 and TP11). With indicator DS4 indicating CLOCKWISE, MOTOR VOLTAGE control T1, when turned from LOW to HIGH, shall cause the voltage of 3.5.1.2 to vary smoothly from zero to its maximum positive value. With indicator DS4 indicating CCW, MOTOR VOLTAGE control T1, when turned from LOW to HIGH, shall cause the voltage of 3.5.1.2 to vary smoothly from zero to its maximum negative value.

3.5.9 POTENTIOMETER switch S6. With DS1 indicating ON, POTENTIOMETER switch S6 shall serve to change indicator DS6 from OFF to ON and the reverse, and to apply or disconnect the voltages of 3.5.1.3 and 3.5.1.4 from the appropriate terminals in accordance with indicator DS6.

3.6 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. The antenna test set shall be soldered per the general requirements and tasks A, C, and D of MIL-STD-2000.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5 and all applicable drawings. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50 deg to +90 deg F.
b.	Relative humidity	90% RH Max.
c.	Pressure	28 to 32 inches of Hg.

4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of one antenna test set for inspection in accordance with 4.3.2.

4.3.2 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection.

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Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. All components, assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table I shall be performed on a 100% basis.

TABLE I. Functional tests
(100% Inspection)

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURE
Critical:		None	N/A
Major:			
101.	Electrical	3.4	4.7.2
102.	Performance	3.5	4.7.3
Minor:		None	N/A

4.5 Special sampling. One antenna test set out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table I and shall then meet the tests in Table II. Unless otherwise specified in the contract the items shall be returned to the lot upon successful completion of testing.

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TABLE II. Special sampling

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURE
Critical:		None	N/A
Major:			
103.	Vibration	3.3.1	4.7.1.1
104.	Shock	3.3.2	4.7.1.2
105.	Altitude	3.3.3	4.7.1.3
106.	Temperature	3.3.4	4.7.1.4
107.	Humidity	3.3.5	4.7.1.5
Minor:		None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product

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may have formulated, furnished, or in any way

which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table III shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

TABLE III. Test equipment list

ITEM NUMBER		DESCRIPTION
1.	Ohmmeter	Up to 1.4 megohms ±2%
2.	Ac voltmeter	Ranges of 0 to 50 volts, 0 to 250 volts, ±3%; sensitivity of 5,000 ohms per volt
3.	Dc voltmeter	Range of 0 to 10 volts, 0 to 50 volts ±2%, sensitivity of 20,000 ohms per volt
4.	Dc ammeter	Range of 0 to 10 amperes ±2%
5.	Test fixture	Per schematic Figure 1
6.	Oscilloscope	Passband dc to 15 MHz, sensitivity of 50 mV per cm

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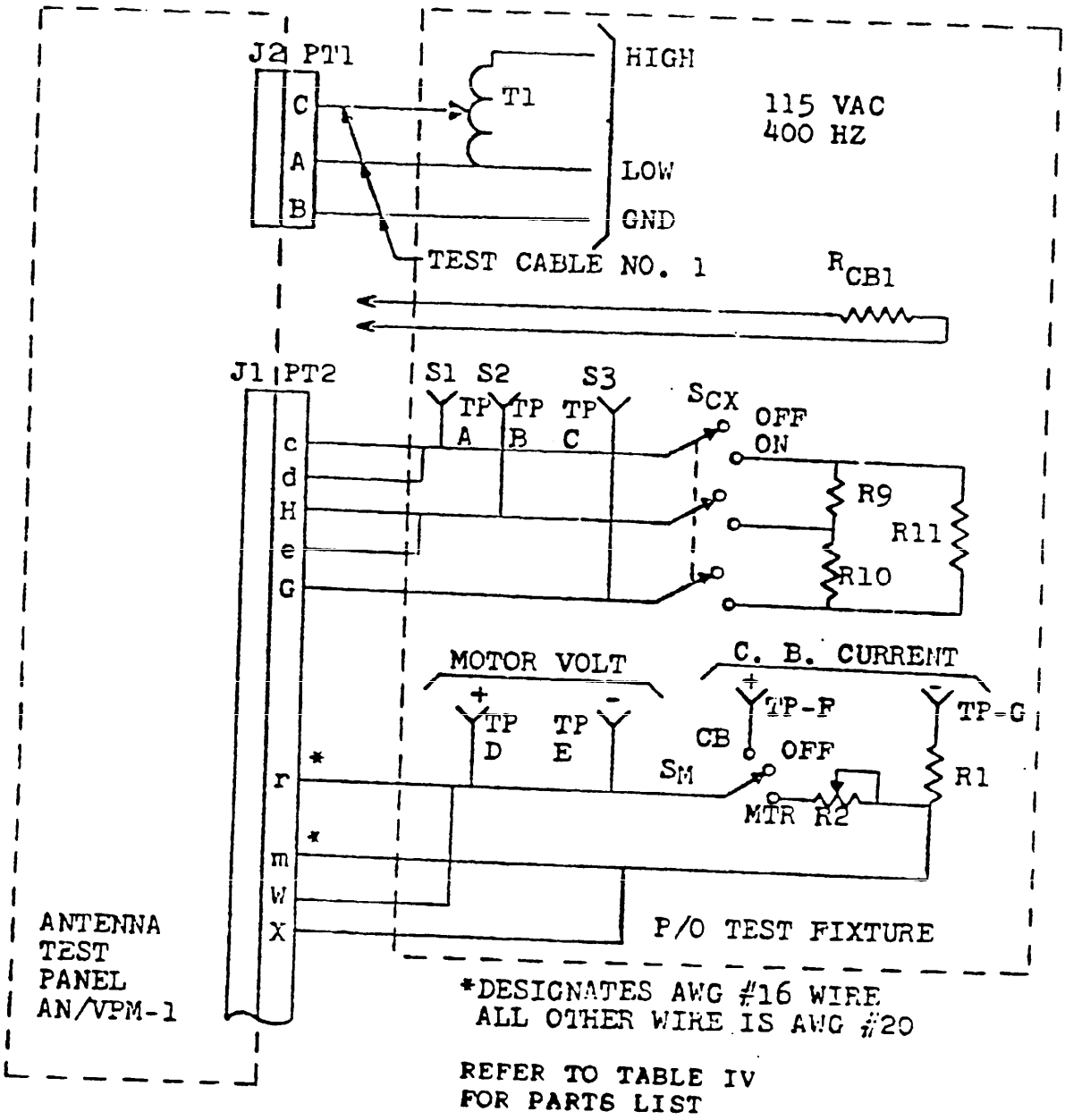


FIGURE 1. Test fixture.

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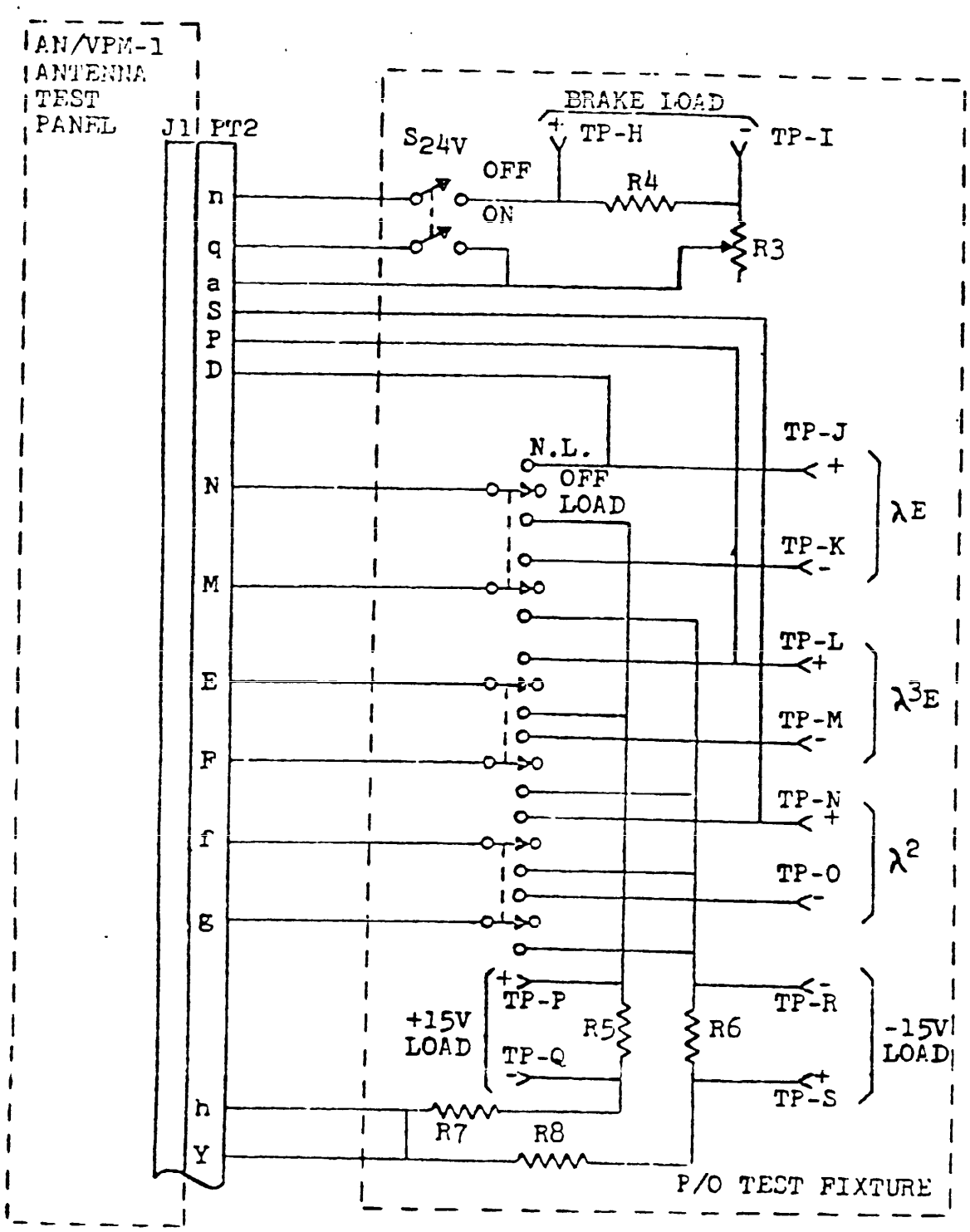


FIGURE 1. Test fixture (cont)

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TABLE IV. Test fixture parts list

REFERENCE DESIGNATION NUMBERS	NOMENCLATURE AND DESCRIPTION
P T1	Connector, Test Cable 1, (MS3126E-12-3S)
P T2	Connector, Test Cable 2, (MS3126E-20-39P)
CB-A	Circuit Breaker, 3A (MS22073-3)
R1	Resistor, Fixed, 1 ohm, 175 W
R2	Resistor, Adj., 2 ohms, 100 W, set at 1.5 ohms
R3	Resistor, Adj., 180 ohms, 5 W, set at 140 ohms
R4	Resistor, Fixed, 10 ohms, 1/2 W
R5	Resistor, Fixed, 330 ohms, 1/2 W
R6	Resistor, Fixed, 39 ohms, 1/2 W
R7	Resistor, Fixed, 3.3 K, 1.2 W
R8	Resistor, Fixed 15 K, 1 W
R9, R10, R11	Resistor, Fixed, 15 K, 1 W
R CB1	Resistor, Var., 5 ohms, 50 W
TP-A, -B, -C, -S	Jack, Banana, Blue
TP-D, -F, -H	Jack, Banana, Red
TP-E, -U	Jack, Banana, Blk
TP-G, -I, -J -L, -P, -Q, -T	Jack, Banana, Yel
TP-K, -M, -N, -O. -R	Jack, Banana, Grn
T1	Transformer, Variable, 120 V, 400 Hz, 3 ampere
S CX	Switch, 4PST, (MS25068-23)
S 24V	Switch, DPDT, (MS35059-23)

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TABLE IV. Test fixture parts list (cont)

REFERENCE DESIGNATION NUMBERS	NOMENCLATURE AND DESCRIPTION	
S , S , 1/ M LE	Switch, DPDT, Center Off,	(MS35059-21)
3 2 1/ SL E, SL		
P (Qty. 2) RCB1	Plug, Banana, Red	

4.7 Methods of Inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810 except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514.3, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only Steps 4 through 8 of paragraph II-2.1 apply. The antenna test set shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the antenna test set shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516.3 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz.

After the application of all the shock impulses, the antenna test set shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

1/ L = greek letter lambda

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4.7.1.3 Altitude (operating and non-operating). This test shall be performed as specified in Procedures II and III, Method 500.2 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501.2 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502.2 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507.2 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Electrical tests shall be conducted by completing the following procedure in order.

4.7.2.1 Continuity. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.1.

4.7.2.2 Resistance. Resistance shall be measured with an ohmmeter (Table III, item 1) for compliance with 3.4.2.

4.7.3 Performance.

4.7.3.1 Switches, indicators, and no-load voltages.

4.7.3.1.1 Set up. Open circuit breakers CB1 and CB2. Connect test cable No. 1 of test fixture, Item 5, and Figure 1, to connector J2 of antenna test set. Turn MOTOR VOLTAGE control T1 to LOW. Plug test fixture in to 115 volt, 400 Hz power. Adjust Input Voltage Control on Item 5 to apply 115 volts to antenna test set. No indicators shall be illuminated.

4.7.3.1.2 Circuit breaker CB1. Depress POWER circuit breaker CB1 and observe that either the OFF or ON part of POWER indicator DS1 is illuminated.

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4.7.3.1.3 POWER switch S1. Depress POWER switch S1 a few times and observe that indicator DS1 changes from OFF to ON and the reverse, and that the OFF or ON part of indicator DS2 and the OFF or ON part of indicator DS6 are illuminated. Leave DS1 ON legend illuminated.

4.7.3.1.4 SYNCHRO EXCITATION switch S2. Depress SYNCHRO EXCITATION switch S2 a few times and observe that indicator DS2 changes from OFF to ON and the reverse. Leave DS2 ON legend illuminated. Use voltmeter (Table III, item 2) to measure 115 ± Vac between SYNCHRO EXCITATION test points CX-R1 and CX-R2 (TP1 and TP2).

4.7.3.1.5 MOTOR TEST switch S3. Depress the switches to produce these results on the indicators.

Switch	Desired Indicator Result
(a) MOTOR TEST (S3)	DS3 not illuminated
(b) BRAKE RELEASED (S5)	DS5 not illuminated
(c) CLOCKWISE-CCW (S4)	DS4 not illuminated
(d) POTENTIOMETER (S6)	DS6 OFF legend illuminated
(e) Any combination of switches	DS7 not illuminated (External connection required to illuminate DS7)

Depress MOTOR TEST switch S3 a few times and observe that the MOTOR TEST part of indicator DS3 is illuminated and extinguished, and the reverse. Leave the MOTOR TEST part of DS3 illuminated.

4.7.3.1.6 BRAKE RELEASED switch S5. Depress BRAKE RELEASED switch S5 and observe that indicator DS5, the ON part of indicator DS3, and either the CLOCKWISE or the CCW part of indicator DS4, are il DS5 illuminated.

4.7.3.1.7 BRAKE VOLTAGE - no load. Connect voltmeter (Table III, item 3) to BRAKE VOLTAGE test points, + (TP12) and - (TP13). Verify BRAKE VOLTAGE indications as specified in 3.5.1.1.

4.7.3.1.8 CLOCKWISE-CCW switch S4. Depress CLOCKWISE-CCW switch a few times and observe that indicator DS4 changes from CLOCKWISE to CCW and the reverse. Leave DS4 CLOCKWISE legend illuminated.

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4.7.3.1.9 MOTOR VOLTAGE - no load. Use voltmeter (Table III, item 3) to measure MOTOR VOLTAGE from + (TP8) to - (TP9). Turn MOTOR VOLTAGE control of T1 slowly from LOW to HIGH. The dc voltage shall increase smoothly from 1 to +19 ±2.0 Vdc. Depress CLOCKWISE-CCW switch S4 so that indicator DS4 indicates CCW. Note that voltage on voltmeter is not disturbed. Turn MOTOR VOLTAGE control to low.

4.7.3.1.10 POTENTIOMETER switch S6. Depress POTENTIOMETER switch S6 a few times and observe that indicator DS6 changes from OFF to ON and the reverse. LEAVE DS6 ON legend illuminated.

4.7.3.1.11 $+KI^2M$ and $-KI^2M$ voltages - no load.

a. Use voltmeter (Table III, item 3) to measure +15 ±1.2 Vdc from $+KI^2M$ (TP17) to RET (TP18), and -15 ±1.2 Vdc from $-KI^2M$ (TP19) to RET (TP18). Record the readings obtained.

b. Connect test cable No. 2 of test fixture (Table III, item 5) to connector J1 of antenna test set. Turn switches S_{LE}, SL^2, and SL^3E [1/] to NL and all other switches on test fixture to OFF.

c. Use voltmeter (Table III, item 3) to measure the following dc voltages. Their values should be the same as recorded in a. above from $+KI^2M$ to RET:

(1) TP-J on test fixture to RET on antenna test set

(2) TP-L on test fixture to RET on antenna test set

(3) LE SIG [1/] to RET on antenna test set

(4) L^3E SIG [1/] to RET on antenna test set

[1/] L = greek letter lambda

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d. Use voltmeter (Table III, item 3) to measure the following dc voltages. Their values should be the same as recorded in a. above from -K1^{2}M to RET:

(1) TP-K on test fixture to RET on antenna test set

(2) TP-M on test fixture to RET on antenna test set

(3) TP-N on test fixture to RET on antenna test set

(4) TP-O on test fixture to RET on antenna test set

(5) L^2 SIG [1/] to RET on antenna test set

4.7.3.1.12 Turn off switches. Set switches for the next step as follows:

Switch	Desired Indicator Result
POTENTIOMETER (S6)	DS6 indicates OFF
CLOCKWISE-CCW (S4)	DS4 indicates CLOCKWISE (will go out on next step)
BRAKE RELEASED (S5)	DS5 not illuminated
MOTOR TEST (S#)	DS3 not illuminated
SYNCHRO EXCITATION (S2)	DS2 indicates OFF (will go out on next step)
POWER (S1)	DS1 indicates OFF

4.7.3.2 Circuit breakers.

4.7.3.2.1 POWER circuit breaker CB1.

a. Set Input Voltage Control of test fixture (Table III, item 5) to 0 Vac. Connect 5 ohm, 50 watt resistor (R_{CB1} of item 5) across CX-R1 and CX-R2. Connect voltmeter (Table III, item 2) across the same test points. Close POWER switch S1 and SYNCHRO EXCITATION switch S2.

[1/] L = greek letter lambda

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b. Increase the voltage from Input Voltage Control and observe that POWER circuit breaker CB1 trips at a voltmeter indication above 7.5 Vac, but less than 15 Vac after an interval not longer than 20 seconds.

c. Open S2 and S1. Remove 5 ohm resistor and reset Input Voltage Control to 115 Vac. Reset CB1.

4.7.3.2.2 MOTOR CURRENT circuit breaker CB2.

a. Turn switch S_M on the test fixture to the CB position. Connect ammeter (Table III, item 4) across test points TP-F and TP-G on the test fixture.

b. On antenna test set, short out MOTOR CURRENT meter M1 and resistor R1. Set switches to give the following indications:

Switch	Desired Indicator Result
POWER (S1)	DS1 indicates ON
MOTOR TEST (S3)	DS3 indicates MOTOR TEST
BRAKE RELEASED (S5)	DS5 illuminated (also illuminates ON part of DS3)
CLOCKWISE-CCW (S4)	DS4 indicates CLOCKWISE

c. Slowly turn MOTOR VOLTAGE control of T1 from LOW towards HIGH and observe that MOTOR CURRENT circuit breaker CB2 open at an ammeter indication above 5 Adc, but less than 10 Adc after an interval not longer than 30 seconds.

d. Turn switch S_M on the test fixture to the OFF position. On antenna test set turn MOTOR VOLTAGE control of T1 to LOW. Turn S1 off. Remove short across M1 and R1. Reset CB2.

4.7.3.3 Voltages at rated load.

4.7.3.3.1 MOTOR VOLTAGE - rated load.

a. Turn switch S_M on the test fixture to the MTR position. Connect 1.5 ohms of adjustable 2 ohm resistor (R_2 of items) across MOTOR VOLTAGE output of antenna test set. Connect voltmeter (Table III, item 2) across test points CX-R1 and CX-R2. Turn S1 on. Set Input Voltage control of test fixture so that voltmeter indication is 103 Vac (the low limit of line voltage).

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b. Connect voltmeter (Table III, item 3) across TP-D(+) and TP-E(-) on test fixture.

c. Turn MOTOR VOLTAGE control of T1 from LOW to HIGH observing current on MOTOR CURRENT meter M1. When control is at HIGH, meter M1 shall indicate at least 4.0 A. The voltmeter indication shall be 1.5 times the current reading, ±1 V.

d. Turn MOTOR VOLTAGE control of T1 to LOW. Depress S4 so that DS4 indicates CCW. Reverse the polarity of the voltmeter leads on TP-D and TP-E of the test fixture. Repeat c. above. Turn control to LOW.

e. Repeat a. above except set Input Voltage Control of test fixture so that voltmeter indication is 127 Vac (the high limit of line voltage).

4.7.3.3.2 BRAKE VOLTAGE - rated load.

a. Connect voltmeter (Table III, item 2) across test points CX-R1 and CX-R2. Set Input Voltage Control of test fixture so that voltmeter indication is 115 Vac.

b. Turn switch S_{24v} on test fixture to ON and observe that LIMIT SWITCH ACTIVATED indicator DS7 is illuminated.

c. Connect voltmeter (Table III, item 3) to measure BRAKE VOLTAGE from + (TP12) to - (TP13). Voltage shall be at least 22 Vdc and not more than 26 Vdc.

d. Connect voltmeter (Table III, item 3) to measure BRAKE LOAD voltage from + (TP-H) to - (TP-I) on test fixture. Voltage shall be 1.45 and 1.75 Vdc.

e. Turn switch S_{24v} on test fixture to OFF.

4.7.3.3.3 $+KI^2M$ and $-KI^2M$ voltages - rated load.

a. Connect voltmeter (Table III, item 2) across test points CX-R1 and CX-R2. Set Input Voltage Control of test fixture so that voltmeter indication is 103 Vac. Turn POTENTIOMETER switch S6 to ON.

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b. Connect voltmeter (Table III, item 3) across test points $+KI^2M$ and RET. Turn switches S_{LE}, SL^2, SL^3E [1/] on test fixture to NO LOAD. Voltmeter indication shall be +15 ±1.2 Vdc. Record voltmeter reading. Turn switches S_{LE}, SL^2, and SL^3E [1/] to LOAD. Voltmeter indication shall not be more than 0.4 V less than that obtained with switches in NO LOAD position. Record reading.

c. Repeat b. above except connect voltmeter (Table III, item 3) across test points $-KI^2M$ and RET. Voltmeter indication shall be -15V ±1.2 Vdc.

d. Repeat a. above except set Input Voltage Control so that voltmeter indication is 127 Vac. Repeat b. and c. above and check voltages under load do not increase more than 0.5 V over those obtained with the input voltage at the lower limit. Record reading.

e. Connect voltmeter (Table III, item 3) across +15V LOAD terminals + (TP-P) and - (TP-Q) of test fixture. Voltage indication shall be 1.24 to 1.48 VDC. Repeat across -15V LOAD terminals, - (TP-R) and + (TP-S). Voltage indication shall be 1.24 to 1.48 Vdc.

f. Open switches S_{LE}, SL^2, and S^3E [1/] on test fixture and observe that the voltages obtained in d. above do not increase more than an additional 0.4 V.

g. Repeat a. above except set Input Voltage Control so that voltmeter indication is 115 Vac. Remove voltmeter. Turn S6 and S1 to OFF.

4.7.3.3.4 SYNCHRO SIGNAL.

a. Short test point CX-R2 (TP1) to test point CX-S3 (TP5). Connect voltmeter (Table III, item 2) across CX-S2 (TP4) and CX-R1 (TP2).

b. Turn on S1 and S2. Voltmeter shall indicate 37 volts (nominal) with synchro at 0 degrees. Remove short from CX-R2 to CX-S3.

c. Connect voltmeter leads to test points TP-A and TP-B on test fixture. Voltage indication shall be 78 ±5 volts. Repeat across TP-B and TP-C and observe same voltage. Repeat across CT-R1 and CR-R2 on antenna test set and observe same voltage.

[1/] L = greek letter lambda

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d. Close switch S_{CX} on test fixture. Connect oscilloscope (Table III, item 6) between CX-S1 (TP3) and CX-S3 (TP5). Using the most sensitive scale on the oscilloscope, the peak-to-peak voltage measurement shall be 150-600 mV. Turn S2 and S1 to OFF. Turn S_{CX} on test fixture to OFF. Disconnect test equipment.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified in paragraph 3.7.

5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with Packaging Data Sheet P10554576.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The antenna test set is used for third and fourth echelon maintenance of the antenna elevation and traverse servo drive assemblies of Radar Set AN/VPS-2. The antenna test set contains equipment to test and troubleshoot both servo drive assemblies and any of their individual electro-mechanical components. The antenna test set also contains features needed to align the control differential transmitter (CDX) of Organizational Maintenance Test Set AN/TPM-23.

6.2 Acquisition requirements. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.
b. Applicable stock number.
c. First Article Sample requirements.
d. Packaging requirements, if other than specified in Section 5.

e. Serialization requirements, if applicable.

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, ATTN: AMSMC-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

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6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (key word) listing.

Antenna Test Set AN/VPM-1
Antenna traverse servo drive assembly
Antenna elevation servo drive assembly
Fire Control System
Radar Set AN/VPS-2
Organizational Maintenance Test Set AN/TPM-23
Vulcan Air Defense System (VADS)

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
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Preparing activity:
Army-AR

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-T-60860A (AR)	2. DOCUMENT DATE (YYMMDD) 16 MAY 1990

3. DOCUMENT TITLE
TEST SET, ANTENNA AN/VPM-1

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME US ARMY ARDEC STANDARDIZATION & SPECIFICATION OFFICE	b. TELEPHONE (Include Area Code) (1) Commercial: (201) 724-667- (2) AUTOVON: 880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S/BLDG 6 PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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